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CHE 321; Spring – 2020; Quiz – 3

100 Points

1. (25 points) Three gas phase reactions have the same reaction rate dependence to the concentration of the reactant.
   a. A → 3 B → ~~Volume increases, pressure~~ decreases $P_1V_1 = P_2V_2$
   b. A → C
   c. 2 A → D → ~~volume~~

   Which of these reactions will exhibit the highest conversion in a PFR? Explain.

10 ~~A~~ C, two molecules of A combine into one molecule of D this decreases the pressure which decreases the volume which decreases the concentration of D

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2. (25 points) Why does the conversion for a gas-phase reaction in a PBR depend on the pressure?

Because the pressure is related to concentration $\frac{mol}{L}$ for gases. The A gas adjusts its position to fill the volume of a container, unlike solids or liquids (incompressible). PV=nRT : pressure is related to volume ↓ volume is related to concentration

as pressure drop is significant

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3. (25 points) Two reactions (one gas phase and one liquid phase) have the same reaction rate dependence to the concentration of the reactant.
   a. A → 1/3 B (Liquid Phase)

   10 b. A → 1/3 B (Gas Phase)

   Which of these reactions will exhibit the highest conversion in a PFR? Explain.

b, because 3 molecules of A will go to 1 molecule of B. This will cause the concentration of the gas, which is related to the pressure to change. The liquid this does not change.

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gas: pressure drop → $C_A\uparrow$ → $X_A\uparrow$

4. (25 points) A reaction rate equation is given by $r_A' = k' P_A$. What are the units of k'?

pressure

$\frac{mol}{Pa\ s\ kg}$

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$$r_A' = \frac{[mol]}{[kg][s]} = k'P_A = k'[atm]$$

$$\Rightarrow k' = \frac{[mol]}{[kg][s][atm]}$$